

Angular HandsOn -1

Objective

To install Angular CLI, create a new Angular project, understand the project structure, generate components, and run the Student Course Portal application successfully.

Software Requirements

- Node.js (LTS 20+)
 - npm
 - Angular CLI v20
 - Visual Studio Code
 - Google Chrome
-

Step 1: Install Angular CLI

Open Command Prompt or PowerShell and execute:

```
npm install -g @angular/cli
```

Verify installation:

```
ng version
```

Step 2: Create Angular Project

Navigate to the folder where you want to create the project.

Execute:

```
ng new student-course-portal --routing --style=css
```

When prompted:

Would you like to enable Server-Side Rendering (SSR) and Static Site Generation (SSG)?

Select:

No

Step 3: Navigate to Project Folder

```
cd student-course-portal
```

Step 4: Open Project in Visual Studio Code

Execute:

code .

Step 5: Explore Project Structure

Study the following files:

- angular.json
 - package.json
 - tsconfig.json
 - tsconfig.app.json
 - src/main.ts
 - src/app/app.ts
 - src/app/app.config.ts
 - src/index.html
-

Step 6: Create notes.txt

Create a file named:

notes.txt

Write the purpose of each file.

angular.json

Contains the Angular project configuration such as build, serve, assets, and environments.

package.json

Contains project information, dependencies, and npm scripts.

tsconfig.json

Main TypeScript compiler configuration.

tsconfig.app.json

TypeScript configuration specific to the Angular application.

main.ts

Entry point of the Angular application.

app.ts

Root component of the Angular application.

app.config.ts

Contains application-wide providers and configuration.

index.html

Main HTML page loaded in the browser.

Step 7: Run the Application

Execute:

ng serve

Open the browser and navigate to:

http://localhost:4200

Verify that the Angular application loads successfully.

Step 8: Build the Project

Execute:

ng build

Observe that the **dist** folder is created.

The compiled application code is located in:

dist/student-course-portal/browser/main-xxxxx.js

Step 9: Budget Configuration

Open:

angular.json

Locate:

budgets

Observe:

- maximumWarning
- maximumError

Purpose

- maximumWarning displays a warning when the bundle size exceeds the specified limit.
 - maximumError stops the build when the bundle size exceeds the specified limit.
-

Task 2 – Generate Components

Execute the following commands one by one.

Generate Header Component

```
ng generate component components/header
```

Generate Home Component

```
ng generate component pages/home
```

Generate Course List Component

```
ng generate component pages/course-list
```

Generate Student Profile Component

```
ng generate component pages/student-profile
```

Generated Files

Each component contains:

- component.ts
 - component.html
 - component.css
 - component.spec.ts
-

Step 10: Modify Header Component

Open:

```
src/app/components/header/header.html
```

Replace the content with:

```
<nav>
  <h2>Student Course Portal</h2>

  <a href="#">Home</a>

  <a href="#">Courses</a>

  <a href="#">Profile</a>
</nav>
```

Step 11: Modify Home Component

Open:

src/app/pages/home/home.html

Replace the content with:

```
<h1>Welcome to Student Course Portal</h1>
```

```
<p>
```

This portal helps students browse courses, enroll, and monitor academic progress.

```
</p>
```

```
<div>
```

```
<p>Courses Available: 12</p>
```

```
<p>Enrolled: 3</p>
```

```
<p>GPA: 3.8</p>
```

```
</div>
```

Step 12: Modify Root Component

Open:

src/app/app.html

Replace all existing content with:

```
<app-header></app-header>
```

```
<app-home></app-home>
```

```
<router-outlet></router-outlet>
```

Step 13: Update app.ts

Import the Header, Home, and RouterOutlet components.

Ensure they are included in the imports array of the @Component decorator.

Step 14: Run the Application

Execute:

```
ng serve
```

Open:

http://localhost:4200

Expected Output

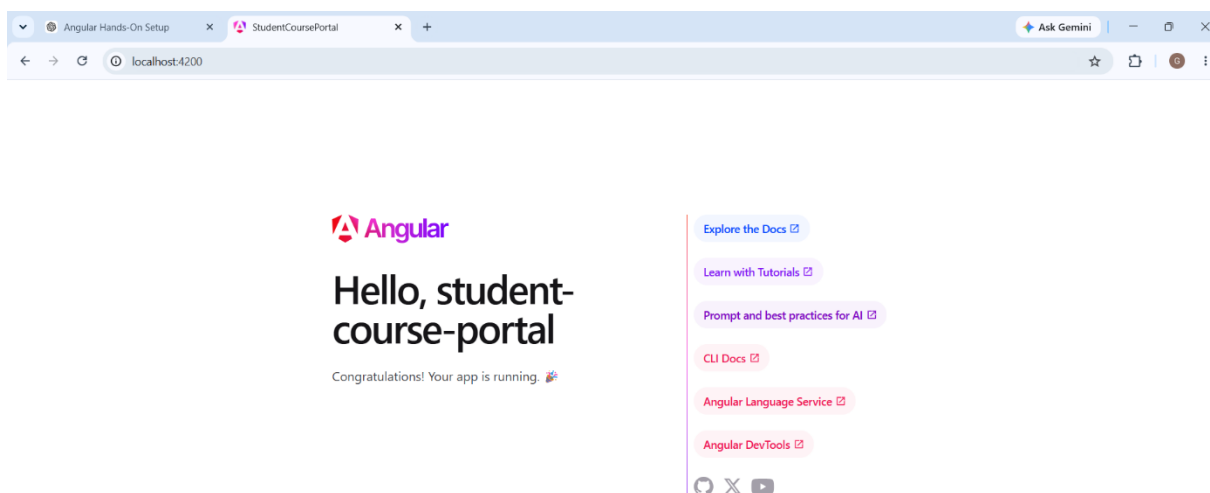
The application displays:

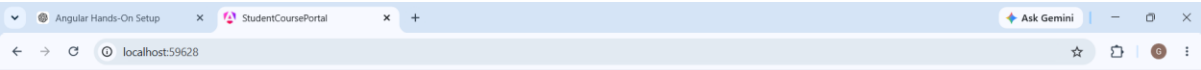
- Student Course Portal navigation bar
- Welcome message
- Description
- Courses Available: 12
- Enrolled: 3
- GPA: 3.8

Conclusion

Successfully installed Angular CLI, created the Student Course Portal project, explored the project structure, generated components, modified the application, and executed the Angular application successfully.

Outputs:





Student Course Portal

[Home](#)[Courses](#)[Profile](#)

Welcome to Student Course Portal

This portal helps students browse courses, enroll, and track academic progress.

Courses Available: 12

Enrolled: 3

GPA: 9.0